



Burning velocities and kinetics of $\text{H}_2/\text{NF}_3/\text{N}_2$, $\text{CH}_4/\text{NF}_3/\text{N}_2$, and $\text{C}_3\text{H}_8/\text{NF}_3/\text{N}_2$ flames



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ABSTRACT

The combustion characteristics and reaction mechanism of mixtures containing nitrogen trifluoride (NF_3) were investigated. Burning velocities for $\text{H}_2/\text{NF}_3/\text{N}_2$, $\text{CH}_4/\text{NF}_3/\text{N}_2$, and $\text{C}_3\text{H}_8/\text{NF}_3/\text{N}_2$ flames were determined for the first time at various equivalence ratios and N_2 mole fractions. The burning velocities of the latter two flames were similar and showed peaks at equivalence ratios of ~ 1.0 , while those of the $\text{H}_2/\text{NF}_3/\text{N}_2$ flames had the pronounced peak at low equivalence ratios where the formation of the wrinkled flames was observed. A detailed kinetic model was constructed to simulate the laminar burning velocities of $\text{H}_2/\text{NF}_3/\text{N}_2$ and $\text{CH}_4/\text{NF}_3/\text{N}_2$ flames. The model accurately reproduced the experimental results. Analyses of the reaction mechanism revealed the major reaction pathways that involve the decomposition of NF_3 , the oxidation and chain-fluorination of H_2 and CH_4 , and the formation of N_2 .

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1. Introduction

Nitrogen trifluoride (NF_3) is the most thermodynamically stable compound among the nitrogen–fluorine species [1] and is widely used as a cleaning and etching agent in the semiconductor and electronics industries [2]. The widespread use of NF_3 for cleaning in chemical vapor deposition processes has rapidly increased the demand for this compound. Because of the strong oxidizing ability of NF_3 , several severe accidents have recently occurred in Japan. To ensure the safe use of NF_3 , assessment of its combustion characteristics is of fundamental importance. Since the oxidizing ability of gases is mainly determined by their oxidation reactions, it is essential to investigate the reaction mechanism for the combustion of mixtures of NF_3 and flammable gases.

Only a limited number of studies have been carried out on the combustion of mixtures involving NF_3 . Croomes [3] investigated the oxidation reaction of n-butane in NF_3 – O_2 mixtures, and found that fluorine atoms react preferentially with hydrogen atoms to produce hydrogen fluoride. Lisochkin and Poznyak [4] reported the flammability limits of mixtures of NF_3 with NH_3 , CCl_3H , CCl_4 , $\text{CF}_2\text{ClCCl}_2\text{H}$, and C_8F_{16} . Although these studies provided some information on the combustion mechanisms and properties of fuel– NF_3 mixtures, there have been no studies carried out with the aim of elucidating of the reaction mechanism. Also, the burning velocity, which is one of the fundamental combustion properties and can

be used to investigate the reaction mechanism, has never been reported for fuel– NF_3 mixtures.

This study aims to obtain the burning velocities of mixtures of NF_3 with H_2 , CH_4 , and C_3H_8 and develop and validate a kinetic mechanism to explore its oxidation reactions. The burning velocities have been measured for $\text{H}_2/\text{NF}_3/\text{N}_2$, $\text{CH}_4/\text{NF}_3/\text{N}_2$, and $\text{C}_3\text{H}_8/\text{NF}_3/\text{N}_2$ flames. A detailed kinetic model was constructed to simulate the laminar burning velocities of $\text{H}_2/\text{NF}_3/\text{N}_2$ and $\text{CH}_4/\text{NF}_3/\text{N}_2$ mixtures, which was used to investigate the reaction pathways and chain-reaction mechanisms for the oxidation of H_2 and CH_4 by NF_3 .

2. Experimental method

The burning velocities were measured in a closed spherical vessel. Figure 1 shows a schematic diagram of the experimental apparatus. A spherical stainless-steel vessel with a volume of 15 L was equipped with gas inlet and outlet ports, electrodes, a pressure transducer, and, in some experiments, an acrylic (PMMA) window. The electrodes were linked to a high-voltage source and generated a spark gap of 1.5–3 mm at the center of the vessel. A strain-gauge-type pressure transducer (Kyowa, PHL-A) was installed on the wall of the vessel and connected to a strain amplifier (Kyowa, CVD-700A).

A sample mixture with a known composition, prepared by the partial pressure method, was introduced into the vessel and ignited by an electrical spark between the electrodes. Pressure profiles after ignition were measured by the pressure transducer,

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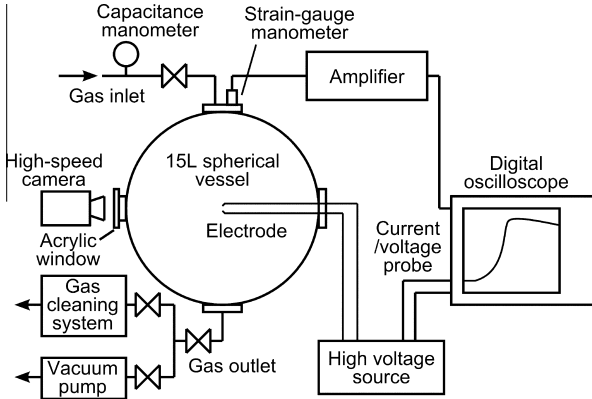


Fig. 1. Schematic diagram of the experimental apparatus.

amplified, and recorded on a digital oscilloscope (Hioki 8861). The gas composition is expressed using the equivalence ratio, ϕ , and the mole fraction of N_2 . The equivalence ratios were defined as $2X(H_2)/3X(NF_3)$, $8X(CH_4)/3X(NF_3)$, and $20X(C_3H_8)/3X(NF_3)$ for the $H_2/NF_3/N_2$, $CH_4/NF_3/N_2$, and $C_3H_8/NF_3/N_2$ mixtures, respectively, where X is the mole fraction of the species in the sample mixture. The initial gas was maintained at room temperature (295 ± 5 K) and the initial pressure was 101.3 kPa. The nominal purity of the gases used was more than 99.9% for NF_3 , H_2 , and CH_4 and more than 99.5% for C_3H_8 .

The burning velocity, S_u , of the sample mixture was determined by the pressure profile method developed by Metghalchi and Keck [5] and Hill and Hung [6]. According to Hill and Hung, the time profile of the mass burned fraction, x , can be expressed as

$$\frac{dx}{dt} = \frac{3S_u}{R} \left\{ 1 - (1-x) \left(\frac{P_0}{P} \right)^{1/\gamma} \right\}^{2/3} \left(\frac{P}{P_0} \right)^{1/\gamma} \quad (1)$$

where R is the inside radius of the vessel, P_0 is the initial pressure of the gas, P is the instantaneous pressure in the vessel, and γ is the specific heat ratio of the unburned gas. The mass burned fraction can be calculated by assuming that it has a linear relationship [6] with the pressure increase, i.e.,

$$x = \frac{P - P_0}{P_e - P_0} \quad (2)$$

Here, P_e is the equilibrium pressure under an adiabatic constant-volume condition. The temperature and pressure dependences of the burning velocity can be expressed as

$$S_u = S_{u0} \left(\frac{T}{T_s} \right)^\alpha \left(\frac{P}{P_s} \right)^\beta \quad (3)$$

where S_{u0} is the burning velocity under the reference condition of $T_s = 298$ K and $P_s = 101.3$ kPa, and α and β are coefficients used to express the temperature and pressure dependences, respectively. Substituting Eqs. (2) and (3) into Eq. (1) and integrating the resultant equation from $t = t_0$ to an arbitrary time t yield

$$P = P_{t=t_0} + \int_{t_0}^t \frac{3S_{u0}(P_e - P_0)}{R} \left[1 - \frac{P_e - P}{P_e - P_0} \left(\frac{P_0}{P} \right)^{1/\gamma} \right]^{2/3} \left(\frac{T}{T_s} \right)^\alpha \left(\frac{P}{P_s} \right)^\beta \left(\frac{P}{P_0} \right)^{1/\gamma} dt \quad (4)$$

Because the unburned gas is assumed to be compressed isentropically and $T_0 \approx T_s$ and $P_0 \approx P_s$, Eq. (4) can be reduced to be

$$P = P_{t=t_0} + \int_{t_0}^t \frac{3S_{u0}(P_e - P_0)}{R} \left[1 - \frac{P_e - P}{P_e - P_0} \left(\frac{P_0}{P} \right)^{1/\gamma} \right]^{2/3} \left(\frac{P}{P_s} \right)^c dt \quad (5)$$

where c represents a pressure–temperature dependence coefficient given as

$$c = \alpha + \beta - \frac{\alpha - 1}{\gamma} \quad (6)$$

The burning velocity, S_{u0} , and the coefficient c were obtained by a least-squares fit of the observed pressure–time profile to Eq. (5). The equilibrium pressure, P_e , was determined by using NASA-CEA2 program [7]. Figure 2 shows pressure profiles following the ignition of the $H_2/NF_3/N_2$, $CH_4/NF_3/N_2$, and $C_3H_8/NF_3/N_2$ mixtures at $X(N_2) = 0.8$ and $\phi = 1.0$. All the profiles could readily be fitted to Eq. (5). The bold curves represent the pressure profiles calculated using Eq. (5) with the fitted parameters.

To assess the accuracy of the pressure profile method, the burning velocities of CH_4 /air mixtures were also measured. They were determined to be 17.8, 37.2, and 27.0 cm s^{-1} at equivalence ratios of 0.7, 1.0, and 1.3, respectively. These values are in close agreement with reported values [8] and references therein.

In some experiments of the $H_2/NF_3/N_2$ and $CH_4/NF_3/N_2$ mixtures, the burning velocity was also determined by directly observing the flame propagation. Light emission from the flame was monitored with a high-speed monochrome video camera (Vision Research, Phantom v7.3) through the PMMA window with a sample rate of 5 kHz and exposure for 0.1 ms. The time evolution of the flame radius was determined by following the emission front in the flame images. This analysis was only performed for flames with a radius of less than 50 mm. For such flames, the total pressure in the vessel remained nearly constant (the actual pressure rise was less than 3%).

The flame speed, V_s , can be evaluated from the measured flame radius as $V_s = dr/dt$. This flame speed corresponds to the stretched one, which is related to the unstretched laminar flame speed [9], V_l , as

$$V_s = V_l - L\kappa \quad (7)$$

where L is the Markstein length and κ is the stretch rate, defined as $\kappa = 2V_s/r$. Then, the unstretched laminar burning velocity, S_l , can be determined as

$$S_l = V_l \frac{\rho_b}{\rho_u} \quad (8)$$

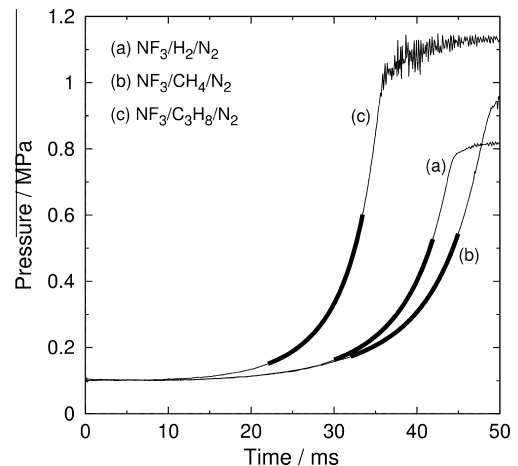


Fig. 2. Time profiles of the pressure following the spark ignition of $NF_3/H_2/N_2$ (a), $NF_3/CH_4/N_2$ (b), and $NF_3/C_3H_8/N_2$ (c) mixtures at $X(N_2) = 0.80$ and $\phi = 1.0$. The bold lines are the profiles calculated using Eq. (5) with the fitted parameters.

Here, ρ_u is the initial density of the unburned gas and ρ_b is the density of the burned gas obtained from an equilibrium calculation at a constant pressure. Examples of the observed flame speeds are plotted in Fig. 3 as a function of the stretch rate. The unstretched flame speeds were obtained by extrapolating the stretch rate to zero by linear least-squares fitting. Because the Markstein lengths obtained in the present study were small ($|L| < 0.01$ cm), nonlinear effects in the extraction of the laminar flame speeds [10] are expected to be negligible.

3. Kinetic modeling

3.1. Simulation method

Kinetic simulations were performed to calculate the burning velocities of the $H_2/NF_3/N_2$ and $CH_4/NF_3/N_2$ flames and investigate the reaction mechanisms. The burning velocities were computed using the Cantera program [11] as laminar flame speeds of one-dimensional freely propagating flames. A detailed kinetic model constructed in the present study was employed for the simulation. The flame computations were performed at an initial temperature of 298 K and an initial pressure of 101.3 kPa with adaptive meshes specified by the maximum relative gradient and curvature between two adjacent points of 0.1. Additional calculations with the gradient and curvature thresholds of 0.01 were also performed; the resultant burning velocities agreed with the results obtained using the thresholds of 0.1 to within 3%. A sensitivity analysis was performed to identify important reaction steps. The sensitivity coefficients for the burning velocities were computed by finite central differences with displacements of the rate constant by $\pm 3\%$ for A factors.

3.2. Model construction

The models of Coppens et al. [12] and Burgess et al. [13] were employed for the C/H/N and C/H/F subsets in the present model, respectively. The rate constants for H/F reactions were taken from kinetic databases [14,15]. For the N/H/F system, kinetic data available from the literature (decomposition of NF_3 , NF_2 , and N_2F_4 [14], $NF_2 + F_2 = NF_3 + F$ [16], $NF_2 + H_2 = HF + NHF$ [17], $NF_2 + H = HF + NF$ [18], $NF_3 + N = NF_2 + NF$ [19], $NF + H = HF + N$ [20], and $NHF + F = HF + NF$ [21]) were adopted as is, and thermochemical data for C/H/F and N/H/F species were taken from a database of CEA2 [7]. Owing to the lack of available data, a large number of

reactions involving N/H/F species were estimated in this study and incorporated into the model.

Part of the rate-constant estimations was carried out using quantum chemical calculations and the transition state theory. The B3LYP/6-311G(d,p) method [22] was used for geometry optimizations and vibrational frequency analyses, and the energies were estimated by the CBS-QB3 [23] method using the Gaussian 03 program [24]. Rate constants were evaluated using the transition state theory with the GPOP program [25]. The rate constants for reactions that proceed without any pronounced barriers were evaluated using the variational transition state theory based on the minimum energy path followed at the B3LYP/6-311G(d,p) level.

3.3. Kinetic model

The dominant reaction steps in the present kinetic model are described as follows

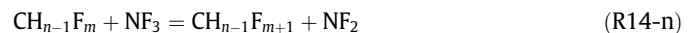
NF_3 degradation



N_2 formation



H_2/CH_4 oxidation



Here, $n = 1, 2, 3$, or 4 and $m = 4 - n$. NF_3 is mainly consumed in the unimolecular decomposition (R1) and F-abstraction reactions by H (R2) and CH_3 (R3) radicals. F atoms produced by the decomposition reaction rapidly react with the fuel molecules, H_2 (R12) and CH_4 (R13). The H/F exchange reaction (R6) also contributes to the degradation of NF_3 , which is followed by reactions (R7) and (R8) to generate NF_2 radicals. The NF_2 radicals formed in these reactions then react with H (R4) and CH_3 (R5) radicals to produce NF radicals.

The recombination reactions of $NF_2 + NF$ (R9) and $NF + NF$ (R10) are important reaction steps for the formation of N_2 . The former reaction proceeds with barrierless association by first forming a chemically activated NF_2NF intermediate with a well depth of 113 kJ mol^{-1} calculated at the CBS-QB3 level relative to the $NF_2 + NF$ asymptote. This intermediate can readily decompose to $N_2F_2 + F$ with a barrier of 31 kJ mol^{-1} relative to the NF_2NF well. The

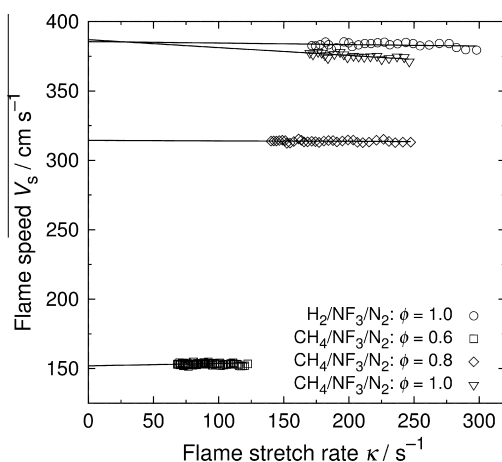


Fig. 3. Observed flame speed as functions of flame stretch rate at $X(N_2) = 0.80$. Lines represent least-squares fits to Eq. (7).

recombination between two NF radicals is also barrierless, forming a N_2F_2 intermediate located 389 kJ mol^{-1} below the reactants. The decomposition of N_2F_2 has a barrier of 106 kJ mol^{-1} and directly produces N_2 and two F atoms. Since the NF_2NF and N_2F_2 wells participating in reactions (R9) and (R10) have low-energy exit channels, the stabilization channels are considered to be negligible for these reactions. N_2F_2 produced in reaction (R9) also rapidly decomposes to $\text{N}_2 + 2\text{F}$.

The oxidation of CH_4 proceeds with a chain-fluoridation mechanism. CH_4 and partially fluoridated methanes are mainly consumed by H-abstraction reactions involving F atoms (R13). The CH_3 and fluoridated methyl radicals produced in these reactions can abstract F atoms from NF_3 (R14) and NF_2 (R15), producing further fluoridated methanes. This cycle gradually replaces hydrogen atoms in CH_4 with fluorine and finally produces CF_4 .

Only selected reactions are presented above and the full kinetic model, which involves 45 species and 251 reactions and includes the kinetic data computed in this study, is available as [Supplementary data](#).

4. Results and discussion

4.1. Experimental results

Figure 4 shows the burning velocities of the $\text{H}_2/\text{NF}_3/\text{N}_2$, $\text{CH}_4/\text{NF}_3/\text{N}_2$, and $\text{C}_3\text{H}_8/\text{NF}_3/\text{N}_2$ mixtures obtained in the present study as functions of ϕ . Open and closed symbols represent the burning velocities determined by the pressure profile and flame visualization methods, respectively. For the $\text{H}_2/\text{NF}_3/\text{N}_2$ flames, the burning velocities obtained by the pressure profile method have a

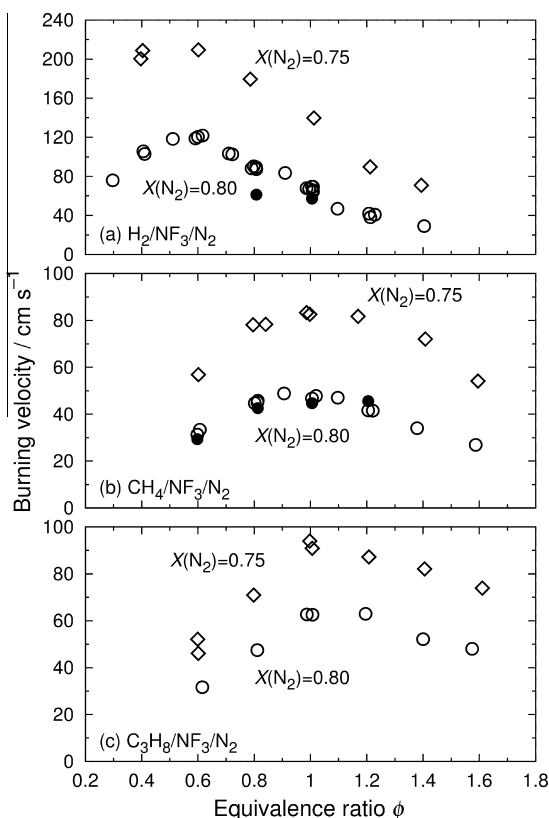


Fig. 4. Burning velocities for the $\text{H}_2/\text{NF}_3/\text{N}_2$ (a), $\text{CH}_4/\text{NF}_3/\text{N}_2$ (b) and $\text{C}_3\text{H}_8/\text{NF}_3/\text{N}_2$ (c) flames. Open circles and diamonds represent the velocities obtained by the pressure profile method at $X(\text{N}_2) = 0.80$ and 0.75 , respectively. Closed circles represent the velocities obtained by the flame visualization method.

moderate peak at approximately $\phi \sim 0.6$, while the $\text{CH}_4/\text{NF}_3/\text{N}_2$ and $\text{C}_3\text{H}_8/\text{NF}_3/\text{N}_2$ mixtures have peaks at $\phi \sim 1.0$. The maximum burning velocities were respectively 122 ($\phi = 0.6$), 48.9 ($\phi = 0.9$), and 62.7 ($\phi = 1.0$) cm s^{-1} for the $\text{H}_2/\text{NF}_3/\text{N}_2$, $\text{CH}_4/\text{NF}_3/\text{N}_2$, and $\text{C}_3\text{H}_8/\text{NF}_3/\text{N}_2$ mixtures at $X(\text{N}_2) = 0.80$, which increased to 209 ($\phi = 0.6$), 83.3 ($\phi = 1.0$), and 94.0 ($\phi = 1.0$) cm s^{-1} at $X(\text{N}_2) = 0.75$. The dependences of the burning velocities on the mole fraction of N_2 at $\phi = 1.0$ are plotted in Fig. 5. The burning velocities for all three flames exhibited a monotonic linear decrease with increasing $X(\text{N}_2)$. The lines in Fig. 5 represent linear fits to the results and are expressed as $S_{u0} = 1237 - 1463 X(\text{N}_2)$, $573 - 657 X(\text{N}_2)$, and $592 - 665 X(\text{N}_2)$ cm s^{-1} for the $\text{H}_2/\text{NF}_3/\text{N}_2$, $\text{CH}_4/\text{NF}_3/\text{N}_2$, and $\text{C}_3\text{H}_8/\text{NF}_3/\text{N}_2$ flames, respectively.

The burning velocities obtained by the flame visualization method, shown as closed circles in Fig. 4, were in good agreement with those obtained by the pressure profile method for the $\text{CH}_4/\text{NF}_3/\text{N}_2$ flames, indicating that the flame curvatures were sufficiently large for the effects of flame stretching on the burning velocity [26] to be negligible in the time windows used in the analysis of the pressure profiles. On the other hand, the flame visualization method gave a significantly lower burning velocity for the $\text{H}_2/\text{NF}_3/\text{N}_2$ flame at $\phi = 0.8$ than that obtained by the pressure profile method. This deviation can be explained by the formation of a wrinkled flame as described below.

4.2. Formation of wrinkled flame

The $\text{H}_2/\text{NF}_3/\text{N}_2$ flames showed the development of wrinkles at the flame surface at low ϕ conditions. Figure 6 shows the time evolution of the emission from the $\text{H}_2/\text{NF}_3/\text{N}_2$ flames at $X(\text{N}_2) = 0.8$ and $\phi = 0.6, 0.8$, and 1.0 . The corresponding flame speeds are plotted in Fig. 7. At $\phi = 1.0$, the flame propagates smoothly throughout the observation period and the flame speed approaches a zero-stretch asymptote. On the other hand, the flame at $\phi = 0.8$ was initially smooth but small wrinkles appeared when it reached a radius of 32 mm at 8 ms . After that, the flame speed rapidly increased as the wrinkled structure developed. Also, the decrease in flame speed in the early stages ($< 5 \text{ ms}$) of propagation indicates a negative Markstein length, which increases the thermal-diffusive instability of the flame [27]. This instability was further intensified at $\phi = 0.6$ for which the unstretched flame speed could no longer be determined.

The unstretched laminar burning velocity at $\phi = 0.8$ plotted in Fig. 4 was calculated from the flame speed at $t < 7 \text{ ms}$ during which

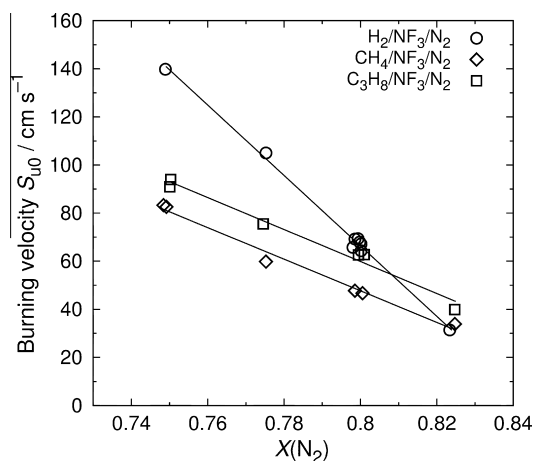


Fig. 5. Burning velocities for the $\text{H}_2/\text{NF}_3/\text{N}_2$ (circles), $\text{CH}_4/\text{NF}_3/\text{N}_2$ (squares), and $\text{C}_3\text{H}_8/\text{NF}_3/\text{N}_2$ (diamonds) flames obtained by the pressure profile method at $\phi = 1.0$ as a function of $X(\text{N}_2)$. Lines represent linear fits to the data.

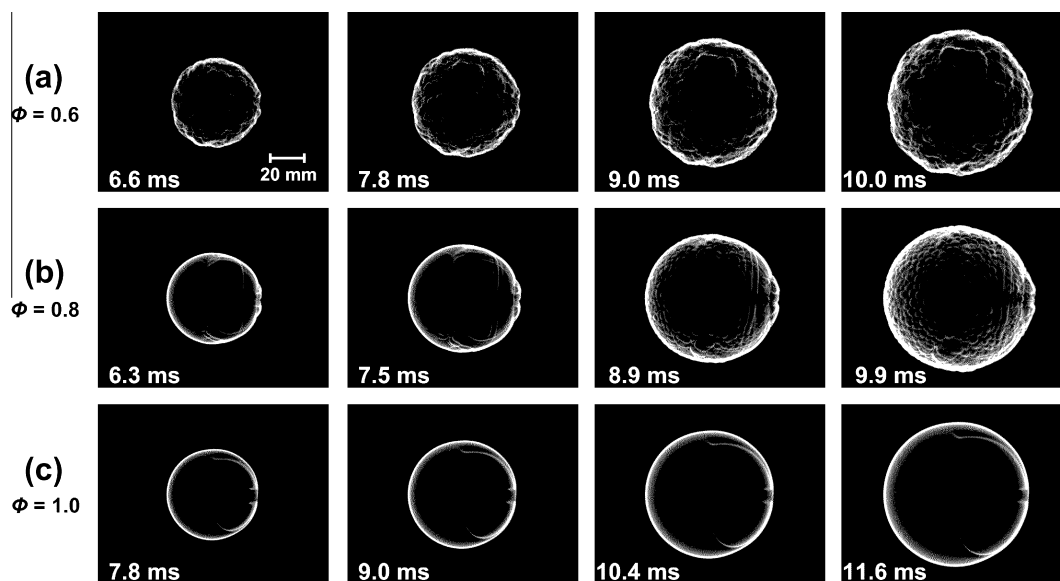


Fig. 6. Images of the flame propagation of the $\text{H}_2/\text{NF}_3/\text{N}_2$ mixtures at $X(\text{N}_2) = 0.80$ and $\phi = 0.6$ (a), 0.8 (b), and 1.0 (c).

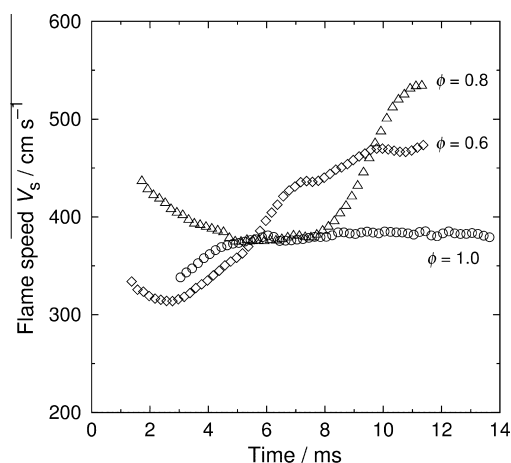


Fig. 7. Time evolution of flame speeds of the $\text{H}_2/\text{NF}_3/\text{N}_2$ mixtures at $X(\text{N}_2) = 0.80$ and $\phi = 0.6$ (diamonds), 0.8 (triangles), and 1.0 (circles).

the flame smoothly propagated. As a result, the pressure profile method significantly overestimated the burning velocity at $\phi = 0.8$ owing to the flame acceleration caused by the instability. The same conclusion can also be drawn for the results of H_2 lean flames ($\phi < 1$), whose Lewis number of the limiting species, H_2 , is substantially less than unity.

4.3. Kinetic simulation

The laminar burning velocities calculated using the kinetic model described in the previous section are plotted and compared with the experimental burning velocities in Fig. 8. The present model accurately reproduced the experimental results for both the $\text{H}_2/\text{NF}_3/\text{N}_2$ and $\text{CH}_4/\text{NF}_3/\text{N}_2$ flames. The discrepancy observed for the $\text{H}_2/\text{NF}_3/\text{N}_2$ flame at low equivalence ratios can be explained by the failure of the pressure profile method employed in this region, in which the flames develop with small wrinkles as discussed above. The simulation results suggested that the burning velocity for the $\text{H}_2/\text{NF}_3/\text{N}_2$ flame has a peak at $\phi = 0.8$.

Sensitivity analyses were performed for the $\text{H}_2/\text{NF}_3/\text{N}_2$ and $\text{CH}_4/\text{NF}_3/\text{N}_2$ flames at $\phi = 1.0$ and $X(\text{N}_2) = 0.75$. The 10 and 16 reactions

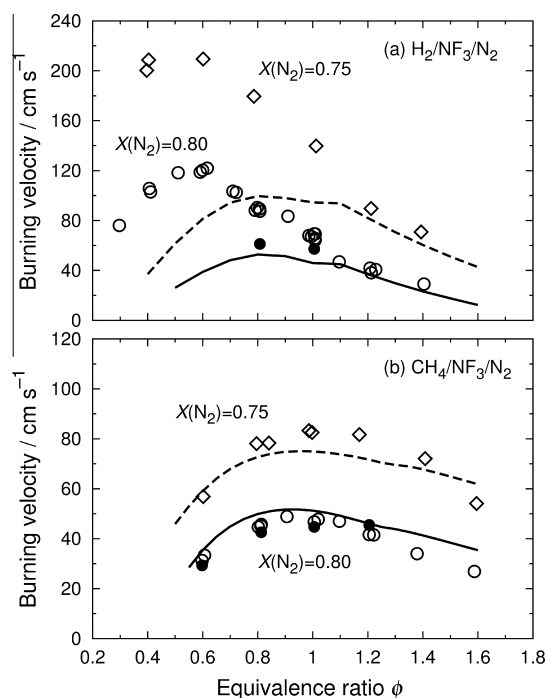


Fig. 8. Comparison of experimental and simulated burning velocities for the $\text{H}_2/\text{NF}_3/\text{N}_2$ (a) and $\text{CH}_4/\text{NF}_3/\text{N}_2$ (b) flames. The symbols denote the experimental burning velocities (see the caption of Fig. 4 for their legends). Solid and dashed lines indicate the computed velocities at $X(\text{N}_2) = 0.80$ and 0.75 , respectively.

with the largest sensitivity coefficients for the burning velocities of the $\text{H}_2/\text{NF}_3/\text{N}_2$ and $\text{CH}_4/\text{NF}_3/\text{N}_2$ flames are shown in Figs. 9 and 10, respectively. The decomposition reaction of NF_3 (R1) has the largest positive sensitivity for both flames. The degradation reactions of NF_2 , (R4) and (R5), as well as NF_3 , (R2) and (R3), are also important since the formation of NF radicals from NF_2 is critically important to ensure the complete consumption of NF_x ($x = 1-3$) species by the recombination reactions of $\text{NF}_2 + \text{NF}$ (R9) and $\text{NF} + \text{NF}$ (R10), both of which finally produce N_2 . For the $\text{CH}_4/\text{NF}_3/\text{N}_2$ flame, the oxidation reactions of CH_4 and fluoromethanes (R13)–(R15) have large positive sensitivities, indicating the importance of the

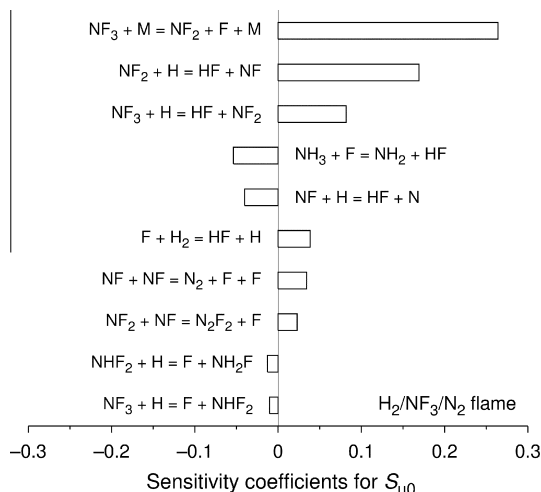


Fig. 9. Sensitivity coefficients for the burning velocity for the $\text{H}_2/\text{NF}_3/\text{N}_2$ flame at $\phi = 1.0$ and $X(\text{N}_2) = 0.75$.

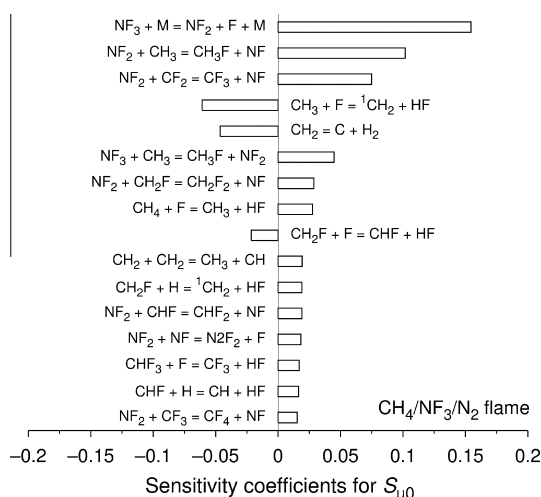


Fig. 10. Sensitivity coefficients for the burning velocity for the $\text{CH}_4/\text{NF}_3/\text{N}_2$ flame at $\phi = 1.0$ and $X(\text{N}_2) = 0.75$.

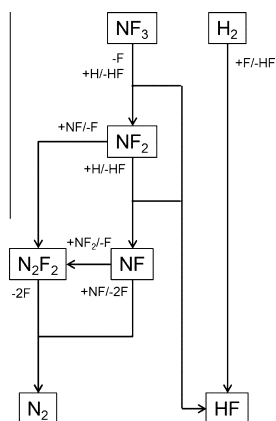


Fig. 11. Schematic diagram of the reaction pathway in the $\text{H}_2/\text{NF}_3/\text{N}_2$ flame.

chain-fluoridation mechanism. Also, the reactions of $^1\text{CH}_2$ and CH_2 had moderate sensitivities for the $\text{CH}_4/\text{NF}_3/\text{N}_2$ flame because the CH_2 produced from the $\text{CH}_3 + \text{F}$ reaction acts as a chain inhibitor.

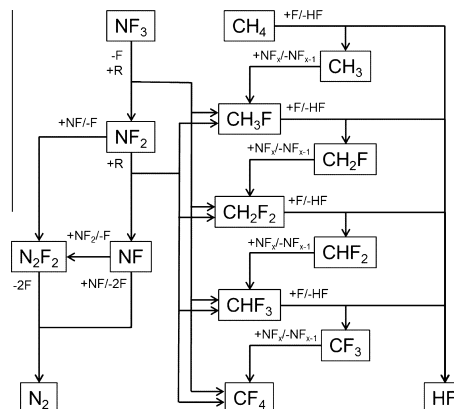
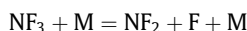


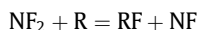
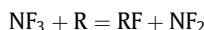
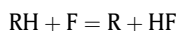
Fig. 12. Schematic diagram of the reaction pathway in the $\text{CH}_4/\text{NF}_3/\text{N}_2$ flame.

To summarize, the chain-reaction mechanism in the $\text{H}_2/\text{NF}_3/\text{N}_2$ and $\text{CH}_4/\text{NF}_3/\text{N}_2$ flames are described by the following reaction steps.

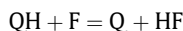
Initiation



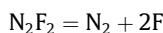
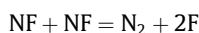
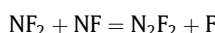
Chain-fluoridation



Inhibition



N_2 formation



Here, R and Q represent alkyl radicals and methylenes, respectively. The major reaction pathways in the $\text{H}_2/\text{NF}_3/\text{N}_2$ and $\text{CH}_4/\text{NF}_3/\text{N}_2$ flames are schematically depicted in Figs. 11 and 12, respectively.

5. Conclusions

Burning velocities for $\text{H}_2/\text{NF}_3/\text{N}_2$, $\text{CH}_4/\text{NF}_3/\text{N}_2$, and $\text{C}_3\text{H}_8/\text{NF}_3/\text{N}_2$ flames were measured for the first time at various equivalence ratios and N_2 mole fractions in a closed vessel by the pressure profile and flame visualization methods. The burning velocities for the latter two flames were similar and exhibited reasonable dependence on the equivalence ratio; both had peaks at $\phi \sim 1$. On the other hand, the $\text{H}_2/\text{NF}_3/\text{N}_2$ flames had peak burning velocities at low equivalence ratio where the development of the wrinkled flames was observed.

A detailed kinetic model was constructed to simulate the laminar burning velocities of $\text{H}_2/\text{NF}_3/\text{N}_2$ and $\text{CH}_4/\text{NF}_3/\text{N}_2$ flames. The model accurately reproduced the experimental results. Analyses of the reaction mechanism revealed that the major reaction pathways for the $\text{H}_2/\text{NF}_3/\text{N}_2$ and $\text{CH}_4/\text{NF}_3/\text{N}_2$ flames involve the decomposition of NF_3 , the oxidation and chain-fluoridation of H_2 and CH_4 by F, NF_3 and NF_2 , and the formation of N_2 from the recombination reactions of $\text{NF}_2 + \text{NF}$ and $\text{NF} + \text{NF}$.

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Appendix A. Supplementary material

Kinetic model for the $\text{H}_2/\text{NF}_3/\text{N}_2$ and $\text{CH}_4/\text{NF}_3/\text{N}_2$ combustion. Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.combustflame.2013.12.001>.

References

- [1] T.M. Klapotke, J. Fluor. Chem. 127 (2006) 679–687.
- [2] A. Tasaka, J. Fluor. Chem. 128 (2007) 296–310.
- [3] E.F. Croomes, Combust. Flame 10 (1966) 71–77.
- [4] Y. Lisochkin, V. Poznyak, Combust. Explos. Shock Waves 43 (2007) 139–142.
- [5] M. Metghalchi, J.C. Keck, Combust. Flame 38 (1980) 143–154.
- [6] P.G. Hill, J. Hung, Combust. Sci. Technol. 60 (1988) 7–30.
- [7] S. Gordon, B.J. McBride, Computer program for calculation of complex chemical equilibrium compositions and applications, NASA Reference, Publication 1311 (1996).
- [8] X.J. Gu, M.Z. Haq, M. Lawes, R. Woolley, Combust. Flame 121 (2000) 41–58.
- [9] P. Clavin, Prog. Energy Combust. Sci. 11 (1985) 1–59.
- [10] A.P. Kelley, C.K. Law, Combust. Flame 156 (2009) 1844–1851.
- [11] D.G. Goodwin, Cantera program 2.0.2, 2013, available at <<http://code.google.com/p/cantera/>>.
- [12] F.H.V. Coppens, J. De Ruyck, A.A. Konnov, Combust. Flame 149 (2007) 409–417.
- [13] D.R. Burgess Jr., M.R. Zachariah, W. Tsang, P.R. Westmoreland, Prog. Energy Combust. Sci. 21 (1996) 453–529.
- [14] D.L. Baulch, J. Duxbury, S.J. Grant, D. Montague, J. Phys. Chem. Ref. Data 10 (1981) 1–721.
- [15] N. Cohen, K.R. Westberg, J. Phys. Chem. Ref. Data 12 (1983) 531–590.
- [16] R.W. Diesen, J. Phys. Chem. 72 (1968) 108–111.
- [17] Y.R. Bedzhanyan, M.A. Ioffe, V.B. Rozenshtein, Y.M. Gershenzon, Kinet. Catal. 31 (1990) 11–16.
- [18] V.B. Rozenshtein, Y.R. Bedzhanyan, Y.M. Gershenzon, Kinet. Catal. 29 (1988) 22–29.
- [19] P.R.P. Barreto, A.F.A. Vilela, R. Gargano, S.S. Ramalho, L.R. Salviano, J. Mol. Struct. (THEOCHEM) 769 (2006) 201–205.
- [20] C.T. Cheah, M.A.A. Clyne, P.D. Whitefield, J. Chem. Soc., Faraday Trans. 76 (1980) 711–728.
- [21] K.B. Hewett, D.W. Setser, J. Phys. Chem. A 101 (1997) 9125–9131.
- [22] P.J. Stephens, F.J. Devlin, C.F. Chabalowski, M.J. Frisch, J. Phys. Chem. 98 (1994) 11623–11627.
- [23] J.A. Montgomery Jr., M.J. Frisch, J.W. Ochterski, G.A. Petersson, J. Chem. Phys. 110 (1999) 2822–2827.
- [24] M.J. Frisch, G.W. Trucks, H.B. Schlegel, G.E. Scuseria, M.A. Robb, J.R. Cheeseman, J.A. Montgomery Jr., T. Vreven, K.N. Kudin, J.C. Burant, J.M. Millam, S.S. Iyengar, J. Tomasi, V. Barone, B. Mennucci, M. Cossi, G. Scalmani, N. Rega, G.A. Petersson, H. Nakatsuji, M. Hada, M. Ehara, K. Toyota, R. Fukuda, J. Hasegawa, M. Ishida, T. Nakajima, Y. Honda, O. Kitao, H. Nakai, M. Klene, X. Li, J.E. Knox, H.P. Hratchian, J.B. Cross, V. Bakken, C. Adamo, J. Jaramillo, R. Gomperts, R.E. Stratmann, O. Yazyev, A.J. Austin, R. Cammi, C. Pomelli, J.W. Ochterski, P.Y. Ayala, K. Morokuma, G. A. Voth, P. Salvador, J.J. Dannenberg, V.G. Zakrzewski, S. Dapprich, A.D. Daniels, M.C. Strain, O. Farkas, D.K. Malick, A.D. Rabuck, K. Raghavachari, J.B. Foresman, J.V. Ortiz, Q. Cui, A.G. Baboul, S. Clifford, J. Cioslowski, B.B. Stefanov, G. Liu, A. Liashenko, P. Piskorz, I. Komaromi, R.L. Martin, D.J. Fox, T. Keith, M.A. Al-Laham, C.Y. Peng, A. Nanayakkara, M. Challacombe, P.M.W. Gill, B. Johnson, W. Chen, M.W. Wong, C. Gonzalez, J.A. Pople, Gaussian 03 Revision E.01, Gaussian Inc., Wallingford CT, 2004.
- [25] A. Miyoshi, GPOP program revision 2010.02.14m3, 2010, available at <<http://www.frad.t.u-tokyo.ac.jp/~miyoshi/gpop/>>.
- [26] Z. Chen, M.P. Burke, Y. Ju, Combust. Theory Model. 13 (2009) 343–364.
- [27] D. Bradley, C.M. Harper, Combust. Flame 99 (1994) 562–572.